

THE WATCHTOWER IMPERATIVE

Lifespan-Calibrated Evaluation for Frontier AI · Working Paper

Casey M. Williams
University of Kansas
Jennifer Victoria Scurrall
ETH Zurich / University of Zurich

THE PROBLEM

AI systems are evaluated against an implicit, cognitively typical adult and deployed to a population where that user is a minority — one structural error behind disparate harms.

- 2021

Voice assistant instructs a 10-year-old to touch a live plug
- 2024

AI companion cultivates dependency; a 14-year-old dies by suicide
- 2023

Insurer algorithm denies post-acute care to elderly enrollees
- 2024

Deepfake video call moves \$25M in one finance transfer

FIVE MODERATORS

- M1

Attribution capacity — models the system's nature & limits
- M2

Self-regulatory capacity — can disengage from reinforcement
- M3

Skill-formation exposure — is the skill still being learned
- M4

Consequence authority — real-world stake committed
- M5

Social substitution risk — AI displacing human ties

LIFESPAN VULNERABILITY MATRIX

Stage	Moderators	Dominant harm
Early childhood (0–7)	M1, M3	Misattributed agency
Middle childhood (8–12)	M1, M3, M5	Offloading in learning
Adolescence (13–17)	M2, M3, M5	Engagement optimization
Emerging adulthood (18–25)	M3, M4	Displaced judgment
Established adulthood (26–64)	M4	Automation bias, peak authority
Later adulthood (65+)	M1,M2,M4,M5	Declining attribution, high authority

U-shaped risk. Vulnerability peaks at both ends of the lifespan.

Authority ≠ scrutiny. Peak decision authority gets the least developmental scrutiny.

FROM CHILD PROTECTION TO LIFESPAN DESIGN

The UK Age Appropriate Design Code, IEEE 2089, and UNCRC General Comment No. 25 encode these commitments — almost exclusively for children. We generalize them into eight extensible Developmental Harm principles: e.g. scale verification friction to consequence authority (DH4); monitor continuously post-deployment (DH8).

THREE TIERS, INDEXED TO THE MATRIX

- 1

Automated Adversarial Simulation
Synthetic personas by moderator profile give population-scale coverage and baselines.
- 2

Expert Human Red Teaming
Panels matched to matrix stage: developmental, human-factors, and geriatric experts alongside security analysts.
- 3

Continuous Post-Deployment Monitoring
Cohort-stratified baselines catch dependency, substitution, skill displacement, and calibration drift over time.

THE GOVERNANCE ASK

Matrix coverage:
the share of matrix cells with an assigned instrument, baseline, and escalation threshold — an auditable metric for regulators, procurement, and insurance underwriting, usable without new statutory categories.

Cognitive vulnerability:
an implicit adult user model, blind to lifespan-varying attribution, self-regulation, and consequence authority.